Greenhouse Controller Design

Status

Section	Status
Requirements	In progress
Plant model	Pending
Control strategy	Pending
Simulation	Pending
Validation	Pending

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1. Requirements

1.1 Scope

Simulation-only control system for a single-zone greenhouse.
Controlled variables: **temperature** and **relative humidity**.
No hardware target; design is language-agnostic at this stage.

1.2 Setpoints

Variable	Setpoint	Tolerance
Temperature (T)	TBD °C	± TBD °C
Relative Humidity (RH)	TBD %	± TBD %

1.3 Sensors

Temperature sensor — Munters P-RTS-2

Parameter	Value
Part number	1212.0.42592 / P-RTS-2
Type	30 kΩ thermistor (2-wire)
Operating range	−40°C to +70°C
Typical accuracy	±0.3°C
Max tolerance at 25°C	±3%
Max cable length	300 m
Min wire size	22 AWG, shielded
Interface	Passive resistive (requires ADC or bridge circuit)

Humidity sensor — Rotem RHS-10 SE

Parameter	Value
Brand / Model	Rotem RHS-10 SE (Plus)
Operating range	−10°C to +70°C
RH range	0–100%
Accuracy	±2% (10–90% RH), ±3.5% (90–100% RH)
Supply voltage	9–12 V AC or DC
Output (selected)	0–3 VDC (linear, 0% → 0 V, 100% → 3 V)
Also available as	4–20 mA or 0–10 VDC
Max cable length	300 m
Interface	Analogue voltage output (requires ADC)

No outside sensors are available. Outside conditions are treated as unknown disturbances.

1.4 Physical Layout

The greenhouse is rectangular. When viewed on a floor plan with north at the top, the **north wall is the long left wall** (the building length runs east–west). The left side pointing north means: standing inside looking east along the length, the north long wall is on your left.

Floor plan (top view):

```
                        N
                        ↑
  ╔═══════════════ M3 (Zijwandbeluchting) ═══════════════╗  ← North wall
  ‖  W                                                E ‖
  ‖ – – – – – – – – – – M2 Dakbeluchting noord – – – – – – – – – ‖  ← ¼ width from N
  ‖                                                         ‖
  ‖  . . . . . . . . . . . . .  ridge  . . . . . . . . . . . . . ‖  ← ½ width
  ‖                                                         ‖
  ‖ – – – – – – – – – – M1 Dakbeluchting Zuid  – – – – – – – – – ‖  ← ¾ width from N
  ‖                                                         ‖
  ╚═══════════════════════════════════════════════════════╝  ← South wall
                        ↓
                        S
       ←———— Length (east–west) ————►
```

Cross-section (looking east along the length, N at left):

```
N wall                  ridge                   S wall
‖                        /\                        ‖
‖   M2 vent            /  \           M1 vent      ‖
‖  [———]             /    \         [———]        ‖
‖         \          /      \       /              ‖
‖═══════════\═══════/═══════\═══/═══════════════‖
‖  [M3 win] ————————————————————————————————  ‖
‖  (north wall, full length)                      ‖
╚═══════════════════════════════════════════════╝
|←— ¼W —→|←———— ¼W ————→|←— ¼W —→|
```

Window positions:

Dimension	Description
Greenhouse orientation	Length east–west; north long wall on the left when facing east
Roof type	Gabled; ridge runs east–west along the full length
M2 vent position on roof	North roof slope, at ¼ of the N–S width from the north wall
M1 vent position on roof	South roof slope, at ¾ of the N–S width from the north wall (¼ from south wall)
M1 and M2 vent extent	Full length of the greenhouse (east to west)
M3 vent position	North wall (long wall), over the full east–west length of that wall

1.5 Actuators

The only actuators are three motorised ventilation windows. There is no heating, cooling, humidification, or dehumidification equipment.

Actuator	Dutch name	Location	Primary effect	Secondary effect
M1	Dakbeluchting Zuid	South roof slope, ¾ width from N, full length	Ventilation (lowers T)	Lowers RH when outside is drier
M2	Dakbeluchting Noord	North roof slope, ¼ width from N, full length	Ventilation (lowers T)	Lowers RH when outside is drier
M3	Zijwandbeluchting	North wall (side wall), full length	Ventilation (lowers T)	Lowers RH when outside is drier

Motor relay box — Hotraco RRK-3

The three motors are driven by a **Hotraco RRK-3** 3-fold window relay box.

Parameter	Value
Model	RRK-3 (3-voudige raamrelaiskast)
Manufacturer	Hotraco Industrial BV
Motor supply	3 × 400 VAC / 50 Hz
Motor power (standard)	3 × 0.37 kW / ±1.2 A
Motor power (maximum)	3 × 0.75 kW / ±2.1 A
Thermal protection	3 × 0.85 / 1.3 A (standard)
Control voltage	24 VAC or 24 VDC
Enclosure	IP54
Dimensions (L × W × H)	220 × 270 × 105 mm
Alarm relay contact	0.5 A / 24 V

Per motor channel, the RRK-3 accepts these control signals (24 V):

Terminal	Signal	Direction
OPEN sturing	Open command	Controller → RRK-3
DICHT sturing	Close command	Controller → RRK-3
COMM. sturing	Common / return	Controller → RRK-3
eindsch. OPEN	End-switch fully open	Motor mechanism → RRK-3
eindsch. DICHT	End-switch fully closed	Motor mechanism → RRK-3
eindsch. NOODSTOP	Emergency stop	Motor mechanism → RRK-3

Critical observation from the wiring diagram: The end-switches connect directly to the RRK-3, not to the controller. The RRK-3 cuts motor power autonomously when an end-switch is triggered. The controller **receives no end-switch signal** and has no way to know that the motor has stopped.

Controller output requirements: 6 × 24 V digital outputs (or relay contacts) — one OPEN and one CLOSE per window motor.

Circuit documentation:

Document	File	Description
Electrical schematic	`documentation/VentilationSystem/ElectriscSchema_001.jpg`	Full electrical schematic — denboer engineering, 12-2-2026, scheme 04, first release
Connection diagram	`documentation/VentilationSystem/AansluitSchema_001.jpg`	Hotraco RRK-3 connection diagram with motor labels (M1 = LUCHING R ZUID, M2 = LUCHING L NORD, M3 = SCHERM/ZIJ)
Full schematic PDF	`documentation/VentilationSystem/Kasventilatiesysteem schema rev 1.pdf`	Greenhouse ventilation system schematic, revision 1

Interface relay — Finder 56.34.8.024.0040

Parameter	Value
Manufacturer part	56.34.8.024.0040
Type	4-pole changeover (4 × wisselcontact)
Control voltage	24 VAC
Contact rating	12 A
Mounting	Plug-in
Enclosure	IP40

These relays serve as the interface between the controller's 24 V digital outputs and the RRK-3 control inputs.

Motor interface:

Each window motor accepts two commands: `OPEN` and `CLOSE`. The motor runs until an end-switch stops it (fully open or fully closed position). The controller has **no feedback** on:

- Current window position (no encoder or potentiometer)
- End-switch activation (no signal reaches the controller)

Implication: The controller cannot modulate window opening proportionally. Each window is effectively a binary actuator: **fully open** or **fully closed**. Intermediate positions are possible by timing the motor, but are unreliable without position feedback.

Decision — partial opening not supported: Timed motor stop to achieve an intermediate window position is explicitly **not implemented**. The reasons are:

- No position feedback is available; a timed stop gives no guarantee of actual opening fraction.
- Repeatability is poor: friction, wear, and temperature affect motor speed over time.
- The control strategy (§ 3) achieves graduated ventilation by selecting how many windows to open, not by modulating individual window position.

Each window is therefore commanded only to its fully-open or fully-closed end position.

Estimated state: The controller must maintain a software-tracked estimated state for each window (`OPEN` / `CLOSED` / `MOVING`). State is inferred from commands issued and elapsed time using a conservative motor run-time estimate (a fixed timeout after which the window is assumed to have reached its end position).

1.6 Control Limitations

Because the only actuators are ventilation windows:

- The system can **only cool** (open windows). It cannot actively heat.
- The system can **only dehumidify** (open windows). It cannot actively humidify.
- Opening windows is beneficial only when outside conditions are more favourable than inside. If outside is hotter or more humid, opening windows makes things worse. Since there are no outside sensors, this is a fundamental uncertainty.
- When T and RH call for conflicting actions (e.g. T too high but RH already too low), a conflict resolution strategy is needed.

1.7 Disturbances

Disturbance	Symbol	Notes
Solar radiation	Q_solar(t)	Primary driver of daytime T rise
Outside temperature	T_out(t)	Unknown to controller; affects ventilation effectiveness
Outside humidity	RH_out(t)	Unknown to controller; affects ventilation effectiveness
Plant transpiration	m_transp(t)	Continuous moisture source; raises RH and slightly raises T

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2. Plant Model

2.1 Overview

The greenhouse is modeled as a single well-mixed zone with two coupled state variables:

- **T(t)**: indoor air temperature [°C]
- **RH(t)**: indoor relative humidity [%]

Coupling: a change in T shifts the saturation vapour pressure, which changes RH even without adding or removing moisture.

2.2 Window Ventilation Model

Each window, when open, creates an air exchange with the outside. The effective ventilation rate depends on window area, wind speed, and stack effect (especially for roof windows). Since none of these are measured, an aggregate effective air change rate is used:

- `ACH_i` — air changes per hour contributed by window i when fully open [h^{-1}], where i ∈ {M1, M2, M3}
- Total ventilation rate: `ACH_total(t) = sum of ACH_i for all open windows`

Roof windows benefit from stack effect (hot air rises), making them more effective for heat removal than the side wall window. Between the two roof windows, M2 (north slope) and M1 (south slope) are assumed to have equal ventilation effectiveness; any asymmetry due to prevailing wind direction is not modelled.

2.3 Temperature Dynamics

First-order thermal model:

```
C * dT/dt = Q_solar(t) - UA * (T - T_out) - ACH_total * V * rho * cp * (T - T_out)
```

Where:

- `C` — thermal capacitance of greenhouse air and contents [J/°C]
- `Q_solar(t)` — solar heat gain through glazing [W]
- `UA` — envelope heat loss coefficient (conduction + infiltration) [W/°C]
- `ACH_total` — total ventilation air change rate from open windows [s^{-1}]
- `V` — greenhouse air volume [m^3]
- `rho` — air density [≈ 1.2 kg/m^3]
- `cp` — specific heat of air [≈ 1005 J/(kg · °C)]
- `T_out` — outside temperature [°C] (disturbance, not measured)

The term `UA * (T - T_out)` captures passive envelope losses/gains.
The term `ACH * V * rho * cp * (T - T_out)` captures active ventilation through open windows.

2.4 Humidity Dynamics

Mass balance on water vapour, using absolute humidity (AH) as the internal state:

```
V * d(AH)/dt = m_transp(t) - ACH_total * V * (AH - AH_out)
```

Where:

- `AH` — indoor absolute humidity [kg/m^3]
- `AH_out` — outside absolute humidity [kg/m^3] (derived from T_out, RH_out; disturbance)
- `m_transp(t)` — plant transpiration rate [kg/s] (disturbance)
- `ACH_total` — ventilation rate from open windows [s^{-1}]

Converting AH to RH using the Magnus formula:

```
p_sat(T) = 610.78 * exp(17.27 * T / (T + 237.3))   [Pa]
AH_sat(T) = 0.622 * p_sat(T) / (p_atm - p_sat(T)) * rho_air
RH = AH / AH_sat(T) * 100                           [%]
```

2.5 Coupling

Opening a window simultaneously:

1. Lowers T (by replacing warm inside air with cooler outside air)
2. Lowers AH (if outside is drier than inside)

3. Changes RH both directly (via AH change) and indirectly (via T change affecting AH_sat)

This tight coupling means T and RH cannot be controlled independently.

2.6 Plant Parameters

Symbol	Description	Value	Unit	Source
C	Thermal capacitance	TBD	J/°C	
UA	Envelope heat loss coefficient	TBD	W/°C	
V	Air volume	TBD	m^3	
ACH_M1	Air change rate for M1 (Dakbeluchting Zuid, south roof slope)	TBD	h^{-1}	
ACH_M2	Air change rate for M2 (Dakbeluchting Noord, north roof slope)	TBD	h^{-1}	
ACH_M3	Air change rate for M3 (Zijwandbeluchting, north wall)	TBD	h^{-1}	
m_transp	Plant transpiration rate	TBD	kg/s	
t_motor	Motor run-time to reach end position	TBD	s	

2.7 Outside Environmental Conditions

For simulation purposes, outside temperature and humidity profiles are derived from historical weather data.

Data source:

- File: `Environment/airTemperature_2025-05-01_to_2025-09-01.csv`
- Period: May 1 to September 1, 2025 (growing season)
- Sampling interval: Approximately 30 minutes
- Total data points: 5538
- Temperature range: −1.8°C to 40.6°C
- Humidity range: 30.3% to 95.7%

Implementation:
A Python module (`Environment/outside_conditions.py`) provides interpolated values for T_out(t) and RH_out(t) at any simulation time. The module supports:

- Linear interpolation between data points
- Cyclic repetition for simulations longer than the dataset
- Direct access functions: `T_out(t)` and `RH_out(t)` where t is elapsed time in seconds
- Extraction of complete day profiles for analysis

Usage in simulation:

```
from Environment.outside_conditions import get_conditions

# Get outside temperature and humidity at time t (seconds)
T_out, RH_out = get_conditions(t)
```

2.8 Sensor Signal Conditioning

The controller must read both sensors via analogue inputs.

Temperature (P-RTS-2 thermistor):

- The 30 kΩ NTC thermistor requires a resistor voltage-divider or Wheatstone bridge to produce a voltage readable by an ADC.
- Resolution needed: ⩽0.1°C → ADC resolution must be sufficient to resolve the non-linear thermistor curve across the relevant operating range.

Humidity (RHS-10 SE, 0–3 VDC output):

- Direct analogue voltage input (0 V = 0% RH, 3 V = 100% RH).
- Sensor accuracy: ±2% RH in the 10–90% range; ±3.5% at extremes.
- The ±2% inherent sensor uncertainty sets the floor on controllable precision — hysteresis bands in the control logic should be wider than this.

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3. Control Strategy

3.1 Constraints

The control strategy must account for:

1. **Binary actuators** — windows are either open or closed (no proportional control)
2. **No position feedback** — estimated state only, updated from commands + motor timeout
3. **No end-switch feedback** — controller cannot confirm when a window has reached its limit
4. **Ventilation-only** — can only cool and dehumidify; cannot heat or humidify
5. **Actuator coupling** — all windows affect both T and RH simultaneously
6. **Unknown outside conditions** — cannot predict whether opening a window will help or harm

3.2 Approach

Given binary actuators and high uncertainty, a **rule-based hysteresis controller** is more appropriate than PID for the actuation layer. PID is unsuitable here because there is no continuous actuator to modulate.

Architecture:

```
[T sensor] ──┐
             ├──► [Control logic] ──► [Window state manager] ──► [Motor commands]
[RH sensor] ─┘                                  ▲
                                     [Estimated window states]
```

Two-layer design:

- **Layer 1 — Decision logic:** Determines the *desired* ventilation level (how many/which windows to open) based on T and RH error and priority rules.
- **Layer 2 — Window state manager:** Tracks estimated state of each window and issues motor commands to reach the desired state. Handles motor timeouts and prevents issuing conflicting commands.

3.3 Decision Logic

Define ventilation demand as an integer level `V_demand ∈ {0, 1, 2, 3}` representing the number of windows to open.

Temperature rule (primary):

```
if T > T_sp + dT_high:   T_demand = 3          (all windows open)
elif T > T_sp + dT_mid:  T_demand = 2
elif T > T_sp + dT_low:  T_demand = 1
elif T < T_sp - dT_hyst: T_demand = 0          (close all, retain heat)
else:                     T_demand = current    (deadband, no change)
```

Humidity rule (secondary):

```
if RH > RH_sp + dRH_high: RH_demand = 3
elif RH > RH_sp + dRH_mid: RH_demand = 2
elif RH > RH_sp + dRH_low: RH_demand = 1
elif RH < RH_sp - dRH_hyst: RH_demand = 0
else:                        RH_demand = current
```

Conflict resolution (T takes priority over RH, as excessive heat is more immediately harmful):

```
V_demand = max(T_demand, RH_demand)
```

Exception: if T is below setpoint and RH is above setpoint, opening windows risks further cooling. In this case a minimum ventilation is applied only if RH exceeds a critical threshold `RH_critical`.

3.4 Window Selection Priority

When `V_demand` windows should be open, the selection order is:

1. **M2 — Dakbeluchting Noord** (north roof slope): highest stack-effect benefit; hot air naturally exits toward the roof ridge and out the north-facing vent.
2. **M1 — Dakbeluchting Zuid** (south roof slope): equal stack-effect to M2; south-facing slope additionally benefits from solar-driven thermal buoyancy during daytime.
3. **M3 — Zijwandbeluchting** (north wall): lowest stack-effect; provides cross-ventilation through the side wall when both roof windows are already open.

3.5 Window State Manager

Each window has an estimated state: `OPEN`, `CLOSED`, or `MOVING`.

```
on command OPEN  window i:
    if estimated_state[i] == CLOSED:
        issue motor OPEN command
        set estimated_state[i] = MOVING
        start timeout timer(t_motor)
    elif estimated_state[i] == MOVING:
        wait (do not issue new command)

on timeout for window i:
    set estimated_state[i] = target_state  (OPEN or CLOSED)

on command CLOSE window i:
    symmetric to above
```

Motor commands are not re-issued while a window is `MOVING` to prevent conflicting signals.

3.6 Hysteresis Parameters (to be tuned)

Parameter	Description	Initial value
T_sp	Temperature setpoint	TBD °C
dT_low	T threshold for 1 window	TBD °C above T_sp
dT_mid	T threshold for 2 windows	TBD °C above T_sp
dT_high	T threshold for 3 windows	TBD °C above T_sp
dT_hyst	T hysteresis band (close threshold)	TBD °C below T_sp
RH_sp	Humidity setpoint	TBD %
dRH_low	RH threshold for 1 window	TBD % above RH_sp
dRH_mid	RH threshold for 2 windows	TBD % above RH_sp
dRH_high	RH threshold for 3 windows	TBD % above RH_sp
dRH_hyst	RH hysteresis band (close threshold)	TBD % below RH_sp
RH_critical	RH level forcing ventilation even when T < T_sp	TBD %
t_motor	Motor run time to reach end position	TBD s
t_min_dwell	Minimum time a window must stay OPEN or CLOSED before a new command is accepted (anti-oscillation)	600 s

3.7 Farmer-Accessible Configuration Parameters

Not all parameters are intended for the farmer to adjust. The table below distinguishes between parameters the farmer can configure to match crop and seasonal needs, and parameters that are fixed at commissioning time by the installer based on physical measurements of the greenhouse and motors.

Farmer-configurable parameters — these express the desired climate and the tolerance around it. They should be revisited when the crop changes or at the start of a new growing season.

Parameter	Plain meaning	Typical range
`T_sp`	Target indoor temperature	18 – 28 °C
`dT_high`	How far above target before all windows open (emergency cooling)	5 – 10 °C
`dT_mid`	How far above target before a second window opens	3 – 6 °C
`dT_low`	How far above target before the first window opens	1 – 4 °C
`dT_hyst`	How far below target before open windows close again	1 – 3 °C
`RH_sp`	Target indoor relative humidity	65 – 80 %
`dRH_high`	How far above target before all windows open (emergency dehumidification)	10 – 20 %
`dRH_mid`	How far above target before a second window opens	5 – 12 %
`dRH_low`	How far above target before the first window opens	3 – 8 %
`dRH_hyst`	How far below target before open windows close again	3 – 8 %
`RH_critical`	Humidity level at which a window is forced open even when the temperature is already below target, to prevent mould	82 – 92 %
`t_min_dwell`	Minimum time (seconds) a window must remain in its current state before it can be commanded to change; prevents rapid open-close cycling when a sensor reading hovers near a threshold	300 – 1200 s

Installer/commissioning parameters — these reflect physical properties of the building and motors. They are set once during installation and should not need to change unless hardware is replaced.

Parameter	Plain meaning	How to determine
`t_motor`	Time for a motor to travel from fully closed to fully open (or back)	Measure on the physical system with a stopwatch
`ACH_M1`, `ACH_M2`, `ACH_M3`	Ventilation capacity of each window when fully open	Derived from window dimensions and airflow measurements or literature values
`V`	Internal air volume of the greenhouse	Calculated from floor dimensions and average roof height
`C`, `UA`	Thermal mass and envelope heat-loss coefficient	Estimated from construction materials or identified from temperature measurements
`m_transp`	Plant water vapour output rate	Measured by water balance or estimated from crop type (see `plantTranspirationRateConsiderations.md`)

Key principle: the farmer interacts only with climate targets and tolerance bands. The controller translates those targets into window commands automatically. Tightening the bands (smaller `dT_low`, `dRH_low`) makes the controller respond sooner but increases window actuation frequency and motor wear. Widening the bands reduces wear but allows larger swings around the setpoint.

3.8 Anti-Oscillation Guard (Minimum Dwell Time)

When a measured value hovers near a control threshold, the hysteresis deadband alone may not prevent rapid open-close cycling. Each window command evaluation happens every `ctrl_interval` (60 s), and if the indoor T or RH stays within a threshold band, the controller can issue alternating OPEN/CLOSE commands faster than the physical system can usefully respond.

Solution — minimum dwell timer (`t_min_dwell`):

After a window settles into `OPEN` or `CLOSED` (i.e., the motor timeout completes), a timer starts. Any command to change that window's state is suppressed until `t_min_dwell` seconds have elapsed since the last settled transition. Suppressed commands are counted for diagnostics.

```
command_window(w, desired):
    if w.is_moving():               return   # motor busy — ignored
    if w.state == desired:          return   # already there — no-op
    if t - w.last_settled < t_min_dwell:
        commands_blocked += 1
        return                               # anti-oscillation guard
    # proceed: issue command, start motor timer
```

Parameter guidance:

Condition	Suggested `t_min_dwell`
Stable climate, low crop load	300 – 600 s
Default (general use)	600 s (10 min)
High humidity risk crop (cucumber)	600 – 900 s
Very stable conditions expected	up to 1200 s

Setting `t_min_dwell` shorter than `t_motor` (motor travel time) is pointless — the motor guard already blocks re-commands while moving. Effective values start at `t_motor + ctrl_interval` and upward.

Interaction with the state machine: the dwell guard operates at the `command_window` layer (§ 3.5). It does not modify the window state machine itself; it simply suppresses the request before it reaches the machine. The state machine transitions are unchanged.

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4. Simulation Plan

4.1 Scenarios

#	Description	Purpose
S1	Daytime solar gain, windows start closed	Validate T control; check window open sequence
S2	High humidity (transpiration), cool day	Validate RH control; check conflict handling
S3	Full 24-hour day/night cycle	Validate disturbance rejection over time
S4	T below setpoint, RH above critical	Validate conflict resolution logic
S5	Motor timeout / stall (window stuck)	Validate state manager robustness

4.2 Performance Metrics

Metric	Definition
Time above T_sp + dT_high	Duration at worst-case overshoot temperature
Time T within tolerance	Fraction of simulation time T is within ± tolerance of T_sp
Time RH within tolerance	Fraction of simulation time RH is within ± tolerance of RH_sp
Window actuation count	Number of open/close commands issued (proxy for motor wear)
Commands blocked by dwell guard	Count of commands suppressed by `t_min_dwell` anti-oscillation guard
Conflict events	Count of T/RH conflict situations triggered

4.3 Simulation Parameters

- Time step: 10 s (Euler integration)
- Simulation duration: 86400 s (24-hour runs for S1–S3); 14400 s (S4); 28800 s (S5)
- Solver: Euler (first-order explicit)
- Outside T and RH profiles: Historical weather data from May–September 2025 (see section 2.7), interpolated via `Environment/outside_conditions.py`
- Implementation: `Simulation/greenhouse_simulation.py` — runs all 5 scenarios; outputs per-scenario CSV and PNG

Usage:

```
python Simulation/greenhouse_simulation.py [S1|S2|S3|S4|S5|ALL]
```

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5. Open Questions

Resolved:

- [x] Sensor types and specifications confirmed (Munters P-RTS-2 thermistor, Rotem RHS-10 SE)
- [x] Motor drive confirmed (Hotraco RRK-3 relay box, 24 V control interface)
- [x] No end-switch feedback to controller confirmed (RRK-3 wiring diagram)
- [x] Controller output interface confirmed (6 × 24 V digital outputs, OPEN + CLOSE per motor)
- [x] Interface relay confirmed (Finder 56.34.8.024.0040, 4-pole, 24 VAC)
- [x] Greenhouse shape and orientation confirmed: rectangular, length east–west, north long wall on left when facing east
- [x] Motor-to-window mapping confirmed: M1 = Dakbeluchting Zuid (south roof), M2 = Dakbeluchting Noord (north roof), M3 = Zijwandbeluchting (north wall)

- [x] Window positions confirmed: roof vents at ¼ and ¾ of N–S width, both running full length; north wall vent running full length of north wall
- [x] Circuit schematic available: `documentation/VentilationSystem/ElectriscSchema_001.jpg` (denboer engineering, 12-2-2026)
- [x] Define outside T and RH profiles for simulation — using historical data from May–September 2025 (`Environment/airTemperature_2025-05-01_to_2025-09-01.csv`), interpolated via Python module
- [x] Choose simulation language/tool — Python; see `Simulation/greenhouse_simulation.py`
- [x] Decide whether partial window opening (timed motor stop) should be supported — **not supported**: no position feedback, poor repeatability; graduated ventilation achieved by opening more windows instead (see § 1.5)

Still open:

- [ ] Confirm setpoints for T and RH
- [ ] Confirm greenhouse dimensions (floor area, height → air volume V)
- [ ] Estimate motor run-time to fully open/close each window (measure on physical system)
- [ ] Estimate plant transpiration rate
- [ ] Confirm controller hardware platform (what will generate the 24 V digital outputs and read analogue inputs)

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6. Revision History

Date	Change
2026-03-05	Initial draft — requirements and design structure
2026-03-05	Major revision — updated to reflect actual hardware: 3 motorised windows, no position/end-switch feedback, ventilation-only actuation, rule-based hysteresis controller replacing PID
2026-03-05	Added hardware specifications from documentation: Munters P-RTS-2 temperature sensor, Rotem RHS-10 SE humidity sensor, Hotraco RRK-3 relay box, Finder 56.34.8.024.0040 interface relay; added sensor signal conditioning notes; updated open questions
2026-03-06	Added greenhouse physical layout (section 1.4): rectangular shape, length east–west, north long wall on left; floor plan and cross-section diagrams; window positions at ¼ and ¾ of N–S width (roof) and full north wall length (side); confirmed motor-to-window mapping M1/M2/M3 with Dutch names; added circuit schematic reference (denboer engineering, 12-2-2026); updated plant model, control priority, and open questions accordingly
2026-03-06	Added outside environmental conditions (section 2.7): historical weather data from May–September 2025 (5538 data points, ~30-min intervals, −1.8–40.6°C, 30.3–95.7% RH); created Python interpolation module (`Environment/outside_conditions.py`) for T_out(t) and RH_out(t) in simulation
2026-03-06	Created Python simulation (`Simulation/greenhouse_simulation.py`): Euler-integrated plant model (§ 2.3–2.4), rule-based hysteresis controller (§ 3), all 5 scenarios (§ 4.1); outside T/RH driven by historical weather data via `OutsideConditions` ; outputs per-scenario CSV and PNG
2026-03-06	Decision recorded: partial window opening (timed motor stop) not supported — no position feedback, poor repeatability; graduated ventilation achieved by opening additional windows (§ 1.5, § 5)
2026-03-06	Added § 3.7 Farmer-Accessible Configuration Parameters: distinguishes farmer-configurable climate targets (T_sp, RH_sp, hysteresis bands, RH_critical) from installer/commissioning parameters (t_motor, ACH, V, C, UA, m_transp); explains trade-off between band width and motor wear
2026-03-06	Added § 3.8 Anti-Oscillation Guard: `t_min_dwell` parameter (default 600 s) suppresses window commands issued before the dwell timer expires; prevents rapid open-close cycling near thresholds; implemented in simulation `command_window()` with `commands_blocked` diagnostic counter